Swapnil Shukla

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Education

JSS Science and Technology University — **Mysuru, Karnataka, India**
Bachelor of Engineering – Computer Science and Engineering; ***CGPA: 8.91*** — *Aug 2021 – Jun 2025*

Jawahar Navodaya Vidyalaya — **Hardoi, Uttar Pradesh, India**
Class 10: ***84.4%****, Class 12:* ***93.6%*** — *Class 10: 2018, Class 12: 2020*

Experience

Samsung R&D Institute Bangalore (SRIB) — **Bengaluru, India**
Software Development Intern – MSVP (Multi-OS Storage Validation Platform) — *Jan 2025 – Jul 2025*

System Validation & QA Automation *(C++20, Python, Shell, Linux, SQL, Spring Boot)*

- Designed and executed **system-level QA test plans** for NVMe/SSD workflows across **Windows, Linux, and macOS** environments.
- Developed **Python and Shell scripts** to automate validation pipelines, log parsing, and regression reporting, improving test coverage and reducing manual effort.
- Troubleshot issues across **storage drivers, kernel logs, backend services, and hardware emulators**, ensuring robust cross-platform reliability.
- Collaborated with QA teams to define test methodologies, track bugs via **Jira**, and enforce product quality standards.

Boost Elimination & Custom C++20 Library Development *(C++20, Boost, Shared Memory, JSON, Multithreading)*

- Replaced heavy **Boost dependencies** with over **1000+ lines of custom C++20 libraries** covering IPC, CLI parsing, JSON handling, and multithreading.
- Built a static **Named Object Registry**, **Custom Allocators**, and **Shared Memory Manager** mirroring **Boost.Interprocess** using modern C++ constructs.
- Improved runtime efficiency, reduced binary size by **500MB**, and enhanced debugging through **GDB**, **Valgrind**, and system log analysis.

Distributed Storage & Virtualization Testing *(C++, Python, KVM)*

- Implemented **distributed workload scenarios** for storage validation, testing concurrent client access and synchronization on MSVP.
- Validated MSVP functionality on **KVM-based virtualized environments**, ensuring compatibility across virtualized deployments.

Projects

Ecomzy: Shopping Web Application

- Developed a fully responsive e-commerce platform with **3+ product categories**, interactive cart flow, and order summary to streamline the shopping experience.
- Optimized frontend assets, Improved load times .

Gupp-Shupp: Real-Time Chat Application

- Engineered a real-time chat system using the **MERN stack** and **Socket.io**, supporting **50+ concurrent users** with seamless live messaging.
- Implemented **JWT-based authentication** to ensure secure user access and reduced unauthorized requests.

Skills

- **Languages:** C++20, C, JavaScript, HTML, CSS, Shell Script, Python, Java
- **Tools & Platforms:** Git, GitHub, GDB, Valgrind, CMake, MSBuild, SWIG, Jira, VS Code, Linux Terminal Commands, Visual Studio, GCC
- **Frameworks & Libraries:** Electron, Node.js, React
- **Databases:** MongoDB, MySQL
- **CS Fundamentals:** Data Structures & Algorithms, OOP, Operating Systems, DBMS, Process Synchronization, Shared Memory, IPC

Achievements

- Solved over **200+ problems** on LeetCode and CodeChef with a **contest rating of 1601**, ranking in the **top 16%**.
- Secured a **Global Rank 4217** in LeetCode Biweekly Contest 134.
- Secured a **Global Rank 654** in CodeChef Starters Contest 142.